Data Analytics for Business

Homework 6 – Part 1

Text Mining

In this homework exercise, we will perform basic text mining analysis utilizing bag of words and latent semantic analysis to classify advertisement posts as relevant or irrelevant. The data set consists of 1,000 ads from a website that serves a specific farming community. The text content contains information extracted from the ad creatives and the ad landing pages. The website owners have labeled each ad as either "relevant" (which are legitimate ads providing relevant information to the community) or "irrelevant" (which are fraudulent, spam, or simply not relevant to the community). Based on these data, they hope to develop a text-mining classifier that can automatically classify new ad posts as relevant or irrelevant. You are hired as a consultant to help them with this task.

Download the data file "**Ads.csv**," which contains the following fields. Complete the tasks outlined below.

Data Description

Column Name	Variable Description
doc_id	Document ID (each document corresponds to an ad)
text	Text content extracted from the ad creatives and the ad landing pages
label	Whether the ad is relevant (1) or irrelevant (0)

Tasks

1. Install and load the three packages that will be needed: **"tm"**, **"SnowballC"**, and **"lsa"**.

2. Import the data into R. Take the following steps:

   - Import the data from the file "**Ads.csv**" using **read.csv()** as usual, and save it to a (data frame) variable (e.g., "mydata");

- Take **the first two columns** of the above data frame (i.e., "doc_id" and "text") and convert it into a corpus using the **DataframeSource()** function in this way: **Corpus**(**DataframeSource**(mydata[,1:2])). Save the corpus to a variable;
- Take **the third column** of the above data frame (i.e., "label"), and save it to a variable. This will be the outcome variable.

3. Perform the following preprocessing of the text:
   - Remove extra white spaces, using **tm_map**(…, **stripWhitespace**);
   - Remove punctuations, using **tm_map**(…, **removePunctuation**);
   - Remove numbers, using **tm_map**(…, **removeNumbers**);
   - Remove stopwords, using **tm_map**(…, **removeWords, stopwords("english")**);
   - Stem the documents, using **tm_map**(…, **stemDocument**);
4. Based on the processed corpus, construct the **term-document matrix** using the **TermDocumentMatrix()** function. Display the first 30 terms in the matrix.
5. Calculate the **TF-IDF** values for the term-document matrix using the **weightTfIdf()** function.
6. Further reduce the dimensionality by performing the **Latent Semantic Analysis (LSA)** and construct **20 concepts**. Use the **lsa()** function and set the **dim** argument to **20**.

   Note: This step is computationally intensive and may take several minutes to complete.

   Once the result is available, convert the "**dk**" component (which stores the **document-concept matrix**) into a data frame.
7. Now we have extracted essential information from the texts and converted them into structured data that are ready for analysis. As we explained in class, what follows is simply the usual classification analysis that we are already very familiar with. Let's perform the logistic regression here.

   First, as usual, separate the data into the training set and the test set. **Set the seed to 1111.** Draw a random sample of 800 documents from the 1,000 documents in the corpus. For the

training and test sets, respectively, combine the corresponding **labels** and the values in the **document-concept matrix** from Step 6.

8. Run a logistic regression with the **label** as the dependent variable and the **document-concept coefficients** as the independent variables using the **glm()** function, just as we did before. Display a **summary** of the logistic regression results. (Inspect the output and recall what they mean.)

9. Evaluate the out-of-sample prediction performance of our model:

    - For ads in the test set, predict the probability of an ad being "relevant" using **predict(…, type="response")**.
    - Use **0.5** as the threshold. Classify the ads with the predicted probability of being relevant greater than or equal to 0.5 as "relevant," and those with the predicted probability less than 0.5 as "irrelevant."
    - Compare the predicted outcomes (i.e., ir/relevancy) with the actual outcomes for all ads in the test set. Construct and display the **confusion matrix**. Calculate the **overall prediction accuracy**. Comment on the performance of our simple text mining analysis.